CONTINUAL SERVICE IMPROVEMENT & METRICS

JÖNKÖPING UNIVERSITY
School of Engineering

Continual Service Improvement (CSI)

- **No IT service is perfect.** On the rare occasion a service is fully aligned with business requirements, it won't stay that way for long. Services need to be continually improved to make sure they stay aligned with changing business needs.
- The concept of Continual improvement **can be applied to IT services, including its structure and its processes.** Service quality, service cost and the efficiency of service delivery all need to be considered as part of any continual improvement initiative.

(Agutter 2013)

WHY USING METRICS?

Why Metrics?

In order to identify improvement opportunities, the measurement of current performance is an important factor. Consider the following sayings about measurements and management:

You cannot manage what you cannot control.
You cannot control what you cannot measure.
You cannot measure what you cannot define.

Conclusion:
Performance metrics are central to all change management

Why metrics?

- Easier to concentrate on what is *important*
- Easier to *spot danger* in time in order to correct it
- Improves *moral* in an organization
- Drives effectiveness
- *Communication* driver
- *Inspires* continuous improvements
- ....

WHAT ARE THE CONSEQUENCES IF NOT USING METRICS?

ITSM Metrics: Consequences if not used

- *Reduced visibility resulting in loss of control*
- *Focus on noise instead of what's important*
- *Reactive fire-fighting mode*
- *Low moral in organization*
- *Cost effectiveness not understood*
- *Customer/employee complaints drive improvements*
- *…*

Measurement and Reporting practise (ITIL)

The purpose of the measurement and reporting practice is to ***support good decision-making by decreasing the levels of uncertainty.*** *This is achieved through the collection of relevant data on various managed objects and the valid assessment of this data in an appropriate context. Managed objects include, but are not limited to, products and services, practices and value chain activities, suppliers and partners, and the organization as a whole.*

ITSM Metrics: Objectives

- To *align* business objectives and value with IT
- To *inform* stakeholders of IT Service Management performance etc.
- To *assist* stakeholders in *understanding* IT performance and issues.
- To *minimize interruption* of the business
- …

ITSM Metrics: Objectives

- *Drive* operational excellence of IT (strategically)
- *Measure* IT process performance
- *Control* IT Service Management processes
- *Maximize* IT *productivity* and performance
- *"Prove"* the *value creation* of the IT organization – how is that even possible?

There are three main types of metric that support CSI:

1. Service metrics

   *measure end-to-end service performance. They will often be built using technology and process metrics. Customers will usually find service metrics easy to understand and feel that they reflect their experience. They might, for example, show **service availability** during agreed service hours.*

2. Process metrics

   *measure how well a service management process is performing. They will include **Critical Success Factors, Key Performance Indicators** and activity metrics, for example, the Incident Management process might be measured on the number of incidents with missing information.*

3. Technology metrics

   *are often associated with component and application-based metrics such as **performance, availability**, etc.' These metrics are normally used within IT and are not as appropriate for customers. They could measure, for example, server performance or CPU utilisation.*

How to design a metric

- A metric should be:
  - *designed in accordance with customer requirements.*
  - *compared to ensure that they are attainable.*
  - *monitored to ensure that they are within the desired limits*

CSF and KPIs

- CSI will also assess the **Critical Success Factors** (CSFs) and Key Performance Indicators (KPIs) in use within an organisation.
- A CSF is ***something that must happen if an IT service, process, plan, project or other activity is to succeed***.
- A Key Performance Indicator (KPI) is *'a metric that is used to help manage an IT service, process, plan, project or other activity.* ***Key Performance Indicators are used to measure the achievement of Critical Success Factors****. Many metrics may be measured, but only the most important of these are defined as Key Performance Indicators and used to actively manage and report on the process, IT service or activity. They should be selected to ensure that efficiency, effectiveness and cost effectiveness are all managed.'*

ITSM Metrics: Definition

- "Measurement is the process of empirical, objective, assignment of numbers to properties of objects or events of the real world in such a way as to describe them.“
- ...*is the assignment of numbers to objects or events according to a rule*.“

- Paper on metrics:

Adobe Acrobat Document

ITSM Metric Challenges

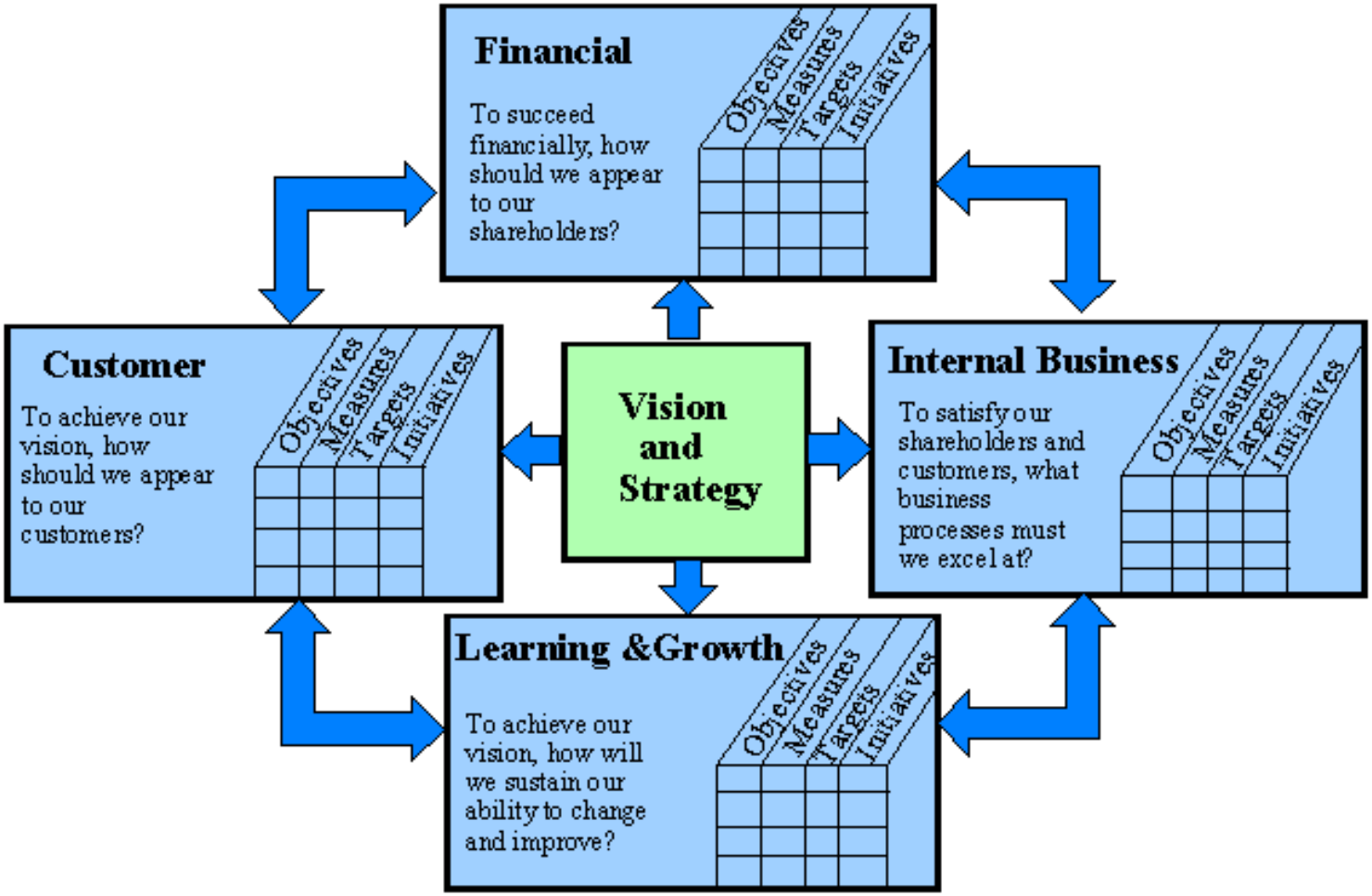
- Metrics not aligned with organizational goals (or conflicts) direct us in the wrong way
- Lack of understanding – misinterpretation
- Wrong level of detail – too much/not enough
- No management sponsorship – commitment
- Lack of education and training
- Stakeholders not identified
- Takes too long to demonstrate benefits

Balanced scorecard (BSC) (Kaplan & Norton)

Four "perspectives“ of BSC:

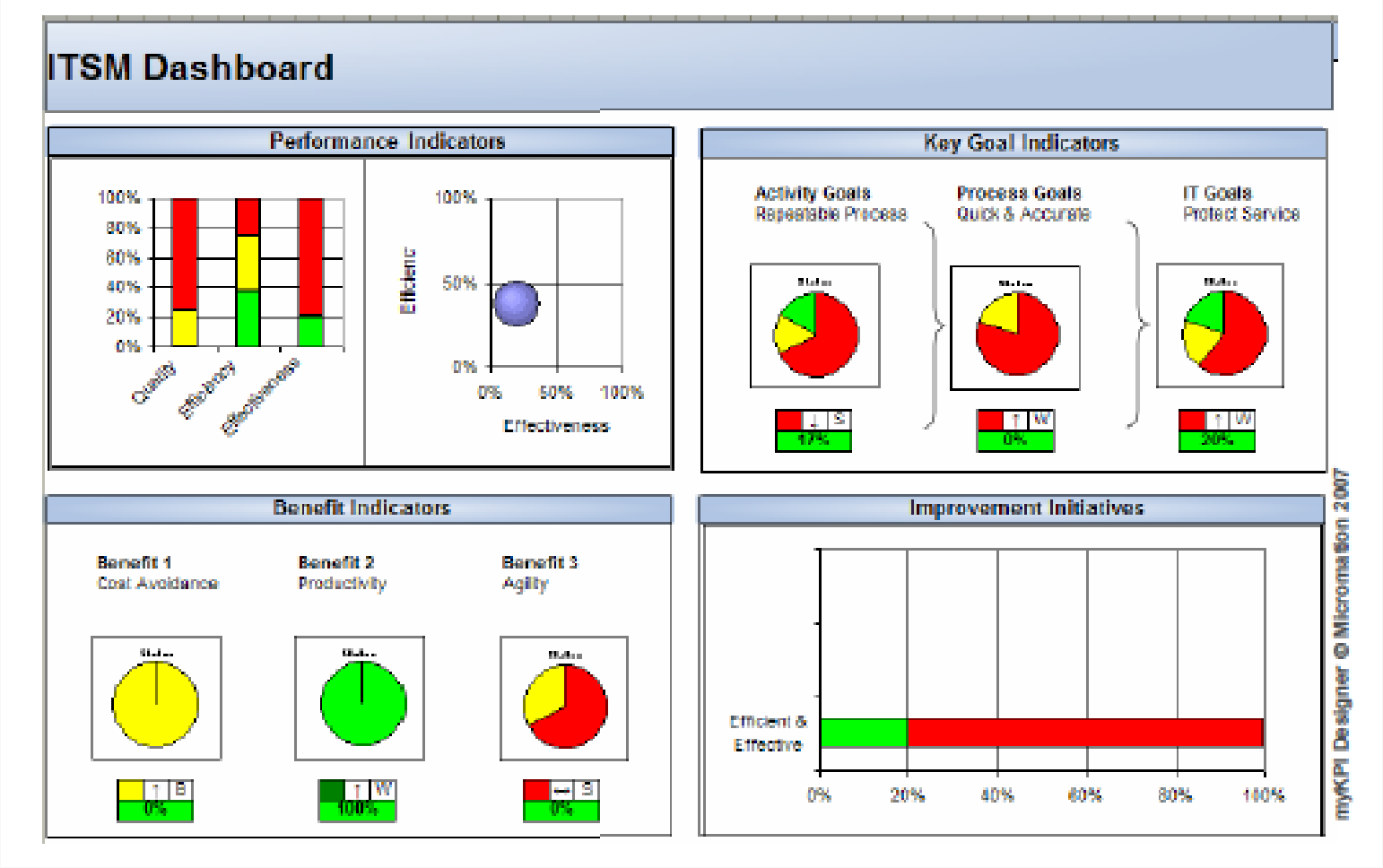
1. **Financial**: encourages the identification of a few relevant *high-level financial measures*. In particular, designers were encouraged to choose measures that helped inform the answer to the question "*How do we look to shareholders*?"
2. **Customer**: encourages the identification of measures that answer the question "*How do customers see us*?"
3. **Internal Business Processes**: encourages the identification of measures that answer the question "*What do we have to be best at, what is important*?"
4. **Learning and Growth**: encourages the identification of measures that answer the question "*How can we continue to improve and enable value*?"

Balanced Scorecard Framework*



* Adapted from Kaplan & Norton 1996. *The Balanced Scorecard*. Harvard Business School Press: 9. Original from HBR Jan/Feb 1996, p. 76.

Visualizing ITSM metrics: Dashboards



KPIs: Examples

- **A business**: the percentage of its income that comes from returning customers.
- **A school**: graduation rates of its students.
- **A Customer Service Department**: percentage of customer calls answered in the first minute.
- **Social service organization**: number of clients assisted during the year.
- **An ITSM department**: Number of errands during a month.

ITSM Metrics: Examples of attributes

Title	Description
ID	ID used to refer to different metrics. (number?)
Metric Title	A short title, which should be as descriptive as possible
ITSM Process	Defines the ITSM processes where the measured value can be used. Could be comma separated list (Service desc, incident mgmt, problem mgmt, change mgmt, release mgmt, config mgmt, SLA, Financial gmt, Capacity mgmt, Continuity mgmt, availability mgmt, security mgmt, Business, project mgmt)
Metric Description	A brief text describing the metric. Should be seen as a complement to Metric Title
Purpose	The purpose of the metric
Relates to business objectives	Describes the relation to one or more of the goals.

ITSM Metrics: Examples of attributes

Title	Description
Target	Describes the result that the business should expect from metric. This should be based on the level of service agreed.
Source	Describes the sources / data the metric is based on.
Specification	Describes exactly what it is that has been measured?
Justification	A description of why this measurement is useful and beneficial for the organization to use
Warning level	Describes when a company should be put worried off and think
Unit	Describes in which or what devices the metric shall be measured in: sek or %
Possible Values	Describes the possible values that the measurement can result in. Ex 1-99999
Audience	An indicator of who the recipient of the measured value results.

ITSM Metrics: Examples of attributes

Title	Description
How to guide	Describes the process (including activities) to reach the result
Formula	Describes the formula
Complexity	Describes how difficult it is to use the metric in a scale of 1-5 with 5 being the hardest.
Metric Category	Describes the type of measurement that exist (KPI, KRI, Business, customer focused etc.
Comment	
Frequency	Is an indication of how often an organization should measure this particular measurement
Quant/Qual	Describes if the measurement is qualitatively or quanitiative
Service Level	Describes what service level measurement has …
Constraints	Describes whether there are any occasions or situations where the metric should not be used

Example of a Customer Satisfaction Survey

ITSM Metrics that could be used in a customer survey…

- AVAILABILITY(scale 1-5)
  - *I believe that the procedures around the incidents works fine.*
  - *I find the time from notification to solution is reasonable.*
  - *I get feedback on submitted cases.*
- COMMITMENT (scale 1-5)
  - *I believe the staff is committed!*
- GENERAL COMMENTS ABOUT AVAILABILITY AND COMMITMENT
  - *Free text.*

ITSM Metrics

- MISSION/ASSIGNMENT (1-5)
  - *I think the mission is carried out to the satisfaction.*
  - *The budget for the mission held.*
  - *The timetable for the mission held.*
  - *Reporting on implementation of the assignment is fine.*
- GENERAL OBSERVATIONS ON THE MISSION
  - *Free text*
- METHODOLOGY
  - *I think the used project methodology is correct and relevant to the assignment*
  - *I think that the development methodology (DSDM, rup, scrum, …) is correct and relevant to the task*
  - *I think the testing methodology is correct and relevant to the assignment*

ITSM Metrics

- GENERAL COMMENTS METHODOLOGY
  - *Text*
- IT SUPPORT IN MISSION IMPLEMENTATION
  - *I think the errand system gives me the support I need.*
  - *I think the Green Hopper gives me the support I need.*
  - *I think ClearQuest gives me the support I need.*
  - *I think that SharePoint gives me the support I need*
  - *I think that ClearCase gives me the support I need*
  - *I think xxx gives me the support I need*
- GENERAL COMMENTS TOOL
  - *Free text*